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SPOTLIGHT STOCK UPDATE Entegris and the Missing Chemical Layer of the AI Economy

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Semiconductors are getting most of the talk these days, and understandably so. Jensen Huang has described the AI buildout as a five-layer cake: energy, chips, infrastructure, models, and applications. That framework is useful because it reminds investors that AI is not just software; it is a physical industrial buildout. But while they are not explicitly named, high-purity chemicals and materials science represent a missing infrastructure layer in four of those five categories, powering the energy transition, enabling chip fabrication, protecting data center infrastructure, and ensuring the reliability of edge applications. This is the quiet but critical foundation of filtration, purification, and contamination-control systems that make the AI economy manufacturable. Before an AI factory can turn power into intelligence, fabs have to turn wafers into usable chips at high enough yields. That is where Entegris matters. This is not a GPU story, and it is not really a chip-design story. It is a materials-science story: the missing layer of the AI stack where again, bits need atoms.

The Missing Layer Beneath Chips

AI is making semiconductor manufacturing harder, not easier. The market talks about more accelerators and data centers, but the deeper bottleneck is increasingly yield. AI accelerators depend on advanced logic, high-bandwidth memory, chiplets, 2.5D and 3D integration, and packaging technologies that push tolerances closer to the physical limit. Each new architecture adds surfaces to clean, layers to planarize, chemicals to deliver, and materials to purify. Because these

specialized chemicals are an evergreen necessity for industrial infrastructure, they remain a structurally PMI-sensitive material that reflects the real-time velocity of the manufacturing cycle. This emphasizes that the five layer cake AI buildout is fundamentally an industrial phenomenon recorded in PMI data, not just a software narrative. In this high-stakes environment, every additional process step creates a fresh opportunity for contamination to destroy value.

Entegris sits directly in that problem set. Its portfolio includes CMP slurries and pads, deposition materials, etch and clean chemistries, filtration and purification systems, fluid-handling products, FOUPs, and other wafer-handling infrastructure. In simple terms, as chips become more complex, Entegris sells more of the products that make that complexity manufacturable. The thesis is not just more wafers. It is more content per wafer, more recurring consumables, and more yield protection as the value of each wafer and package rises.

Intelligence as a Manufactured Commodity

As the market has been realizing that AI is a manufacturing/complexity story rather than a pure software play, the valuation of the “materials layer” changes and especially relative to software. This shifts the AI narrative from a battle of algorithms to a battle of industrial throughput; AI demand doesn’t just show up in cloud margins, it shows up in global manufacturing PMIs, where Entegris’ chemicals act as the high-velocity fuel for the cleanroom. Entegris becomes a proxy for the industrialization of intelligence. As HBM4 and 2nm logic come online in 2026, the scarcity is no longer the design, it is the yield. By controlling the chemicals and the purity of the environment, Entegris effectively taxes every successful wafer start. It is the unglamorous, recurring, and essential infrastructure that ensures the “bits” actually make it into the “atoms” of the hardware.

The CHIPS Act Signal: Materials Are Strategic Infrastructure

This is why the CHIPS Act angle deserves more emphasis. Most investors think about the CHIPS Act through fabs: TSMC in Arizona, Intel in Ohio, Samsung in Texas, and the effort to bring leading-edge manufacturing back to the United States. But Entegris was not an afterthought. In December 2024, Entegris and the U.S. Department of Commerce finalized an award agreement for up to \$77 million to support the company’s Colorado Springs manufacturing center. Entegris said it was the first supplier to semiconductor manufacturers to finalize CHIPS Act award terms. The benefits of this award are moving from the balance sheet to the P&L as the Colorado facility begins initial operations this year, a strategic window that has widened significantly as TSMC’s Arizona commitment has ballooned into a \$165 billion multi-fab cluster with localized advanced packaging.

That detail validates the exact investment thesis. The government was not just funding fabs; it was funding the supplier layer that makes fabs viable. The Colorado Springs center will support liquid filtration products and FOUPs, specialized wafer carriers that protect wafers in manufacturing. Entegris also noted that all of its FOUPs were then manufactured in Asia. That makes the facility strategically important: it localizes a critical purity and handling layer as the U.S. localizes semiconductor production.

The Earnings Progression: From Purity Demand to Content Acceleration

The most important part of the Entegris story is not one quarter of fundamental data. It is the progression in management commentary. In Q1 2025, the company was still dealing with a mixed semiconductor backdrop, but the language was pointing in the right direction. Management highlighted strong demand for CMP consumables and micro contamination-control solutions, while saying customer roadmaps required new materials and ever-greater purity to improve performance and yield.

By Q2 2025, the commentary became more specific. Entegris said growth was driven by unit-driven solutions, particularly CMP consumables, selective etch, and deposition materials. Management also explicitly called out AI-enabled applications as driving significant growth in advanced logic and HBM, even while other fab activity remained subdued. That distinction is important. It showed that the AI-linked portion of the semiconductor supply chain was already behaving differently from the broader cyclical environment.

By Q3 2025, the tone shifted again. Dave Reeder had taken over as CEO, and Entegris highlighted key wins in products critical to advanced nodes, including liquid filtration and purification, deposition materials, and CMP consumables. More

importantly, Reeder framed the story around content per wafer: as devices become more complex, materials science and purity become increasingly critical to performance and yield.

The Q4 2025 commentary crystallized the inflection. Management said 2025 unit-driven growth came from ongoing strength in liquid filtration, selective etch, and CMP consumables, with volume increasing in the most advanced technologies. Reeder then tied the story directly to AI, saying customers are adopting more complex device architectures to support the rapid growth of AI, creating a broad set of opportunities for Entegris. The message moved from purity matters to AI complexity is driving higher content per wafer. That is the kind of progression investors want to see.

HBM, NPUs, and Advanced Packaging: The Multiplier Effect

HBM is the clearest example of why this progression matters. It is not just another memory product; it is a vertically stacked, packaging-intensive architecture that transforms memory from a commodity business into a high-precision chemical challenge. While an HBM wafer displaces three standard DRAM wafers in fab capacity, it generates significantly higher revenue for Entegris because it acts as a process-intensity multiplier. It adds a massive volume of Through-Silicon Vias (TSVs), wafer thinning, and hybrid bonding steps—each requiring the CMP slurries, selective etch chemistries, and ultra-high-purity filtration that Entegris dominates.

As the industry enters the HBM4 era in mid-2026, the transition to 16-layer stacks and hybrid bonding (direct copper-to-copper fusing) creates a “purity wall.” In this regime, even a nanoscale contaminant can cause a catastrophic failure in a high-value package. This makes Entegris’s filtration and cleaning solutions more than a “cost of doing business”—they are essential yield insurance.

The same dynamic applies to NPUs, custom ASICs, and the broader localized buildout. The announcement last week that TSMC has broken ground on its first advanced packaging facility in Arizona—part of its ballooning \$165 billion cluster—validates this shift. By bringing CoWoS and 3D-IC capabilities to the U.S., the industry is localizing the most chemical-intensive part of the AI stack. These architectures depend on silicon interposers and dense connections where CMP must create extreme flatness and cleans must remove residues without damaging delicate structures. The AI architecture changes, but the need for Entegris’ materials and purity stack compounds with every layer of complexity.

Why This Matters Now: The Structural Convergence

Entegris is moving from a cyclical recovery story to a structural AI-complexity story. While it remains exposed to the broader semiconductor cycle, the company’s internal mix has fundamentally shifted. The highest-quality parts of the portfolio are now tethered to advanced-node volume, HBM4 ramp-ups, and 3D-packaging intensity rather than just broad industry headcount. This creates a unique investment setup for the rest of 2026:

- **The Architectural “Common Denominator”:** Entegris is a rare agnostic AI play. Whether the next wave of compute is dominated by Nvidia GPUs, hyperscaler custom ASICs, or the “Small Language Model” shift toward edge NPUs, the physics of the manufacturing process remain the same. Every path requires more precision, more purity, and more process control.
- **The 2nm/HBM4 Yield Threshold:** As we move toward 2nm logic, yield becomes the only metric that matters. When a single defect can destroy a \$30,000 package, the “Purity Tax” Entegris collects becomes a mandatory insurance policy for the entire industry.
- **Localized Alpha:** With the TSMC Arizona Mega-Cluster moving into production and localized advanced packaging, Entegris’s decision to anchor its Manufacturing Center of Excellence in Colorado Springs looks prophetic. Proximity to the “industrial heart” of the domestic AI buildout reduces supply chain friction and solidifies their position as the primary partner for the U.S. “Terafab” movement.

The upcoming April 30, 2026 earnings report is the immediate checkpoint. Investors should look past headline revenue and focus on Materials Solutions margins and management’s commentary on HBM4 content. If Entegris proves it is capturing more value per wafer as complexity rises, it confirms they are the essential infrastructure of the “bits to atoms” transition.

The Bottom Line: Where AI Scarcity Meets Materials Science

The AI economy is becoming a manufacturing-complexity economy. While the market remains fixated on the “visible layers” of the stack, power, chips, applications and models, it is the hidden chemical layer that determines whether the whole system can actually scale. Entegris is the anchor of that layer.

The progression is clear: The CHIPS Act award validates the strategic necessity of a domestic materials supply chain. Management’s earnings commentary shows a steady shift from cyclical recovery to AI-driven content growth. HBM4 and advanced packaging provide the process-intensity multiplier that makes Entegris a “purity tax” on every successful wafer start.

Entegris is not the flashiest name in the AI trade, but it is one of the cleanest ways to own the physical bottleneck of the new industrial regime. By solving for purity and yield at the atomic level, they have become the essential infrastructure for an economy where bits finally need atoms to be useful. In a manufacturing-driven cycle, this is the ultimate proxy for the industrialization of intelligence.

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